

Name: _____



Problem solving with non-linear relationships

Task A: Problem solving with geometric sequences

1) Passing information is another thing which can form a geometric sequence. The bus is going to be late back from a school trip. One pupil phones their parent. The parent phones 5 other parents. Each of those parents phone 5 other parents. Assume it takes 10 minutes to make 5 phone calls.

Time	People
10 mins	5
20 mins	25

a) How many people would know after 40 mins?

(plus the one parent
who initially knew)

b) Why might this not be realistic?

c) How many people would know after 2 hours?

d) Why might this not be realistic?



Task B: Adding sequences

1) In each question, the sequences were created by adding the terms of two of the sequences given below. Work out which two sequences were added in each question.

A 1, 2, 4, 8, 16 ...	B 4, 12, 36, 108, ...	C 4, 9, 14, 19, ...	D 1, 3, 5, 7, 9 ...
E 4, 7, 12, 19, ...	F 1, 6, 16, 31, 51, ...	G 4, 6, 6, 4, 0 ...	H 1, 8, 27, 64, ...

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|---------------------------|----------------------------|-----------------------------|
| a) 5, 12, 19, 26, 33, ... | d) 2, 9, 21, 38, 60, ... | g) 5, 14, 40, 127, 356, ... |
| b) 5, 11, 18, 27, 40, ... | e) 2, 5, 9, 15, 25, ... | h) 8, 15, 20, 23, 24, ... |
| c) 5, 13, 28, 50, 79, ... | f) 2, 11, 32, 71, 134, ... | i) 8, 13, 18, 23, 28, ... |

2) What is the n^{th} term rule for sequence a). How does that relate to the two sequences you have chosen for your answer to a)?

3) By using the examples previously and any of your own examples, investigate the following questions and write a sentence about any patterns you have noticed.

- a) What happens when you add two geometric sequences together?
- b) What happens when you add two quadratic sequences together?
(square and triangle numbers are examples of quadratic sequences)
- c) What happens when you add a linear sequence to a geometric sequence?
- d) What happens when you add a linear sequence to a quadratic sequence?